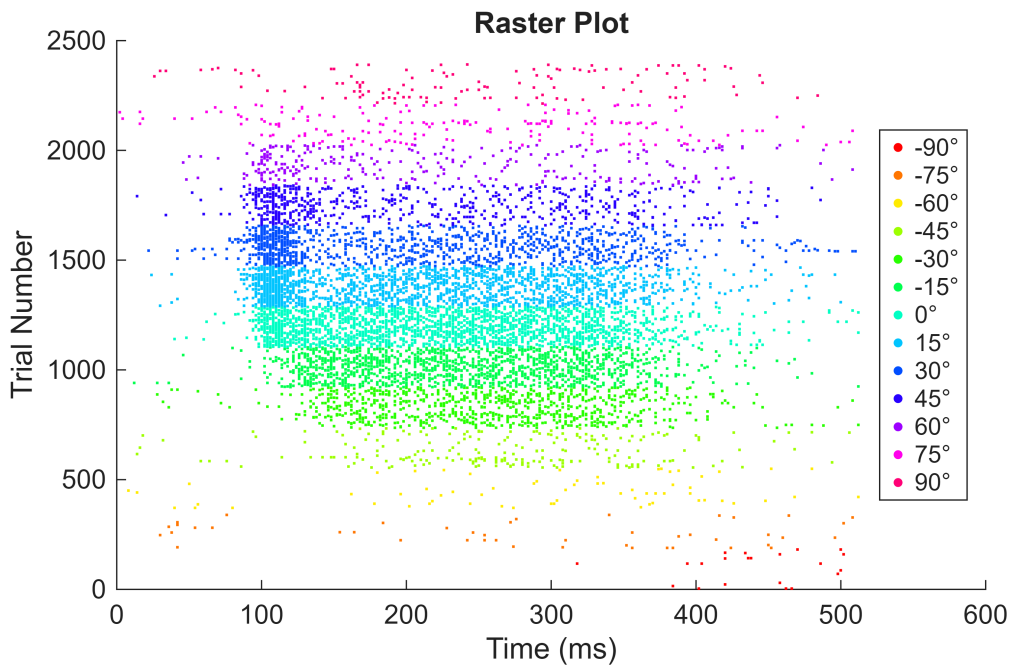
Problem Set 3

Problem 1

```
numTimes = size(mtNeuron.data, 1);
numDirections = size(mtNeuron.data, 2);
numTrials = size(mtNeuron.data, 3);

colors = hsv(numDirections);
directions = linspace(-90, 90, numDirections);

figure; hold on;
trial_offset = 0;
legend_handles = gobjects(numDirections,1);

for direction = 1:numDirections
    for trial = 1:numTrials
        % convert to ms
        spikes = find(mtNeuron.data(:, direction, trial)) * 2;
        y = (trial_offset + trial) * ones(size(spikes));

        plot(spikes, y, '.', 'Color', colors(direction, :), 'MarkerSize', 1);
    end

    legend_handles(direction) = plot(nan, nan, '.', ...
        'Color', colors(direction, :), 'MarkerSize', 10);

    trial_offset = trial_offset + numTrials;
end

xlabel('Time (ms)');
ylabel('Trial Number');
title('Raster Plot');
legend(legend_handles, string(directions) + "°", 'Location', 'best');
```

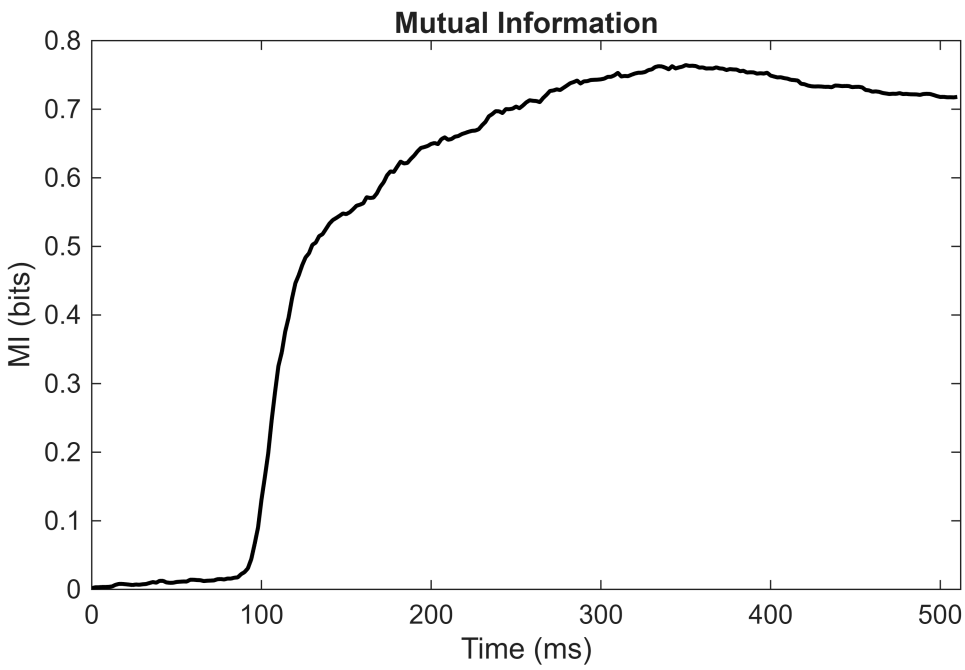


The neuron is directionally tuned and responds most strongly to -30, -45, 0, 15, 30, and 45 degrees.

Problem 2

```
cumulative_spikes = cumsum(mtNeuron.data, 1);
num_directions = size(mtNeuron.data, 2);
num_trials = size(mtNeuron.data, 3);
num_time = size(mtNeuron.data, 1);
P_theta = 1 / num_directions;
time_ms = (0:num_time-1) * 2;
% I(T) = I(T_n, theta) = sum_n P_T(n)log_s(P_T(n|theta)/P_T(n))

mutual_information = zeros(num_time, 1);
for t = 1:num_time
    counts = squeeze(cumulative_spikes(t, :, :));
    count_vals = 0:max(counts(:));
    num_vals   = length(count_vals);

    % P(n | theta)
    Pn_given_theta = zeros(num_directions, num_vals);
    for d = 1:num_directions
        for ni = 1:num_vals
            Pn_given_theta(d, ni) = mean(counts(d, :) == count_vals(ni));
        end
    end

    % P(n), avg
    Pn = mean(Pn_given_theta, 1);
    mi_t = 0;
    for d = 1:num_directions
```

```
        Pn_d = Pn_given_theta(d, :);
        mask = Pn_d > 0 & Pn > 0;
        mi_t = mi_t + P_theta * sum(Pn_d(mask) .* log2(Pn_d(mask) ./ Pn(mask)));
    end
    mutual_information(t) = mi_t;
end

figure;
plot(time_ms, mutual_information, 'k-', 'LineWidth', 1.5);
xlabel('Time (ms)');
ylabel('MI (bits)');
title('Mutual Information');
xlim([0 512]);
```



Before the neuron responds, the cumulative spike count is very close to zero. We can see this in the time before 100ms on our plot where it looks mostly flat. This means that for every trial regardless of direction, the neuron hasn't fired so therefore mutual information will be 0. The rapid increase is when the neuron starts firing. Trials of the neuron's preferred direction cause a larger increase in cumulative spike count, while trials not of the neuron's preferred direction cause the cumulative spike count to stay around 0. This means that there is a greater difference between P(n) and P(n|theta), which causes the mutual information to increase as well. Eventually, this mutual information curve plateaus as the new spikes that fire do not contribute significantly more new information and aren't directionally tuned.

Problem 3

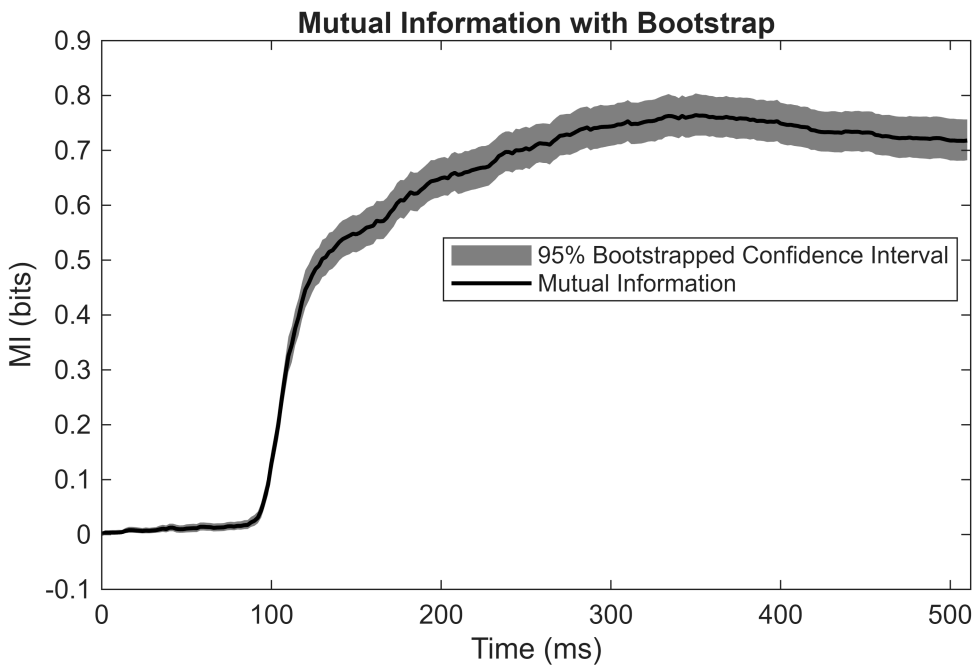
```
n_boot = 1000;
MI_boot = zeros(num_time, n_boot);

for b = 1:n_boot
```

```
    boot_spikes = zeros(num_time, num_directions, num_trials);
    for d = 1:num_directions
        idx = randi(num_trials, 1, num_trials);
        boot_spikes(:, d, :) = cumulative_spikes(:, d, idx);
    end

    MI_raw_boot = zeros(num_time, 1);
    for t = 1:num_time
        counts = squeeze(boot_spikes(t, :, :));
        count_vals = 0:max(counts(:));
        num_vals   = length(count_vals);

        % P(n | theta)
        Pn_given_theta = zeros(num_directions, num_vals);
        for d = 1:num_directions
            for ni = 1:num_vals
                Pn_given_theta(d, ni) = mean(counts(d, :) == count_vals(ni));
            end
        end

        % P(n), avg
        Pn = mean(Pn_given_theta, 1);
        mi_t = 0;
        for d = 1:num_directions
            Pn_d = Pn_given_theta(d, :);
            mask = Pn_d > 0 & Pn > 0;
            mi_t = mi_t + P_theta * sum(Pn_d(mask) .* log2(Pn_d(mask) ./ Pn(mask)));
        end
        MI_boot(t, b) = mi_t;
    end

end

MI_boot_mean = mean(MI_boot, 2);
MI_boot_centered = MI_boot - MI_boot_mean + mutual_information(:);

MI_low  = prctile(MI_boot_centered, 2.5,  2);
MI_high = prctile(MI_boot_centered, 97.5, 2);
```

We need to resample within each direction condition and not pool all the trials together before bootstrapping. This is to preserve the P(n|theta) conditional probability.

```
figure;
fill([time_ms, fliplr(time_ms)], ...
     [MI_low', fliplr(MI_high')], ...
     [0 0 0], 'EdgeColor', 'none', 'FaceAlpha', 0.5);
hold on;
plot(time_ms, mutual_information, 'k-', 'LineWidth', 1.5);
xlabel('Time (ms)');
ylabel('MI (bits)');
```

```
title('Mutual Information with Bootstrap');
xlim([0 512]);
legend('95% Bootstrapped Confidence Interval', 'Mutual Information', 'Location',
'best');
```



Problem 4

To find the latency, we used the first time bin (after three consecutive time bins to reduce the chance of picking up noise) where the mutual information was greater than two standard deviations of the pre-stimulus mean. I set the this pre-stimulus period to be the first 60 seconds, so the first 30 time bins based on the fact that the data is sampled at 2ms.

```
pre_bins    = 1:30; %60 ms
baseline_mu = mean(mutual_information(pre_bins));
baseline_sd = std(mutual_information(pre_bins));
threshold   = baseline_mu + 2 * baseline_sd;

consecutive_bins = 3;
above    = mutual_information > threshold;
latency_bin = NaN;
for t = 1:(num_time - consecutive_bins + 1)
    if all(above(t:t+consecutive_bins-1))
        latency_bin = t;
        break;
    end
end
latency_ms = (latency_bin - 1) * 2;
fprintf('Estimated latency: %d ms (bin %d)\n', latency_ms, latency_bin);
```

```
Estimated latency: 80 ms (bin 41)
```

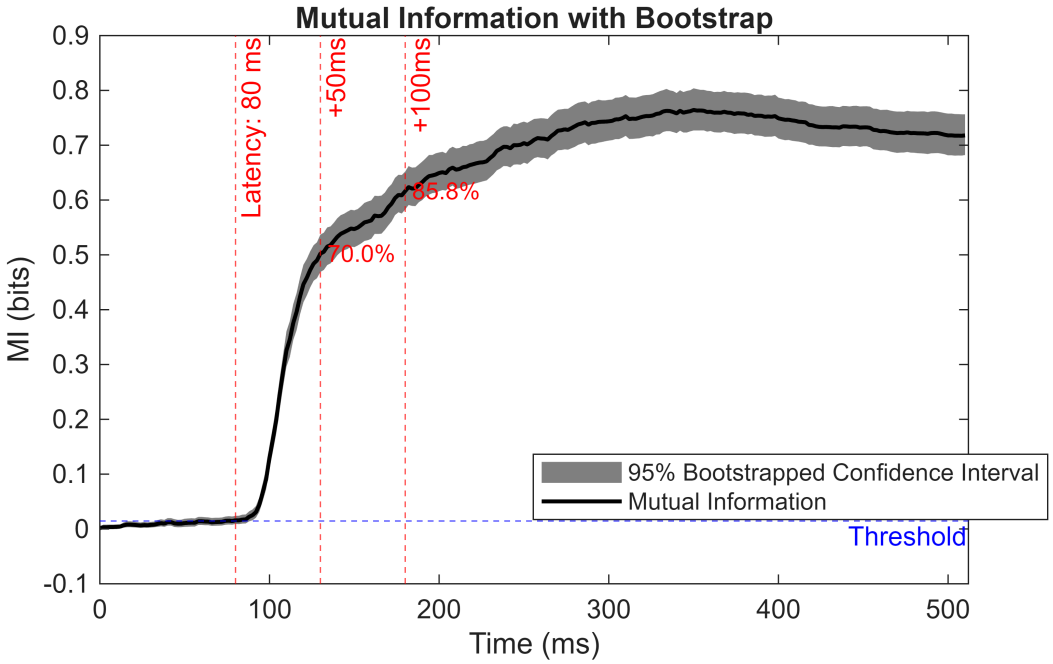
```
total_mutual_information = mutual_information(end);

% 50 and 100ms after latency
bin_50  = latency_bin + round(50  / 2);
bin_100 = latency_bin + round(100 / 2);
bin_50  = min(bin_50,  num_time);
bin_100 = min(bin_100, num_time);

MI_at_50  = mutual_information(bin_50);
MI_at_100 = mutual_information(bin_100);

frac_50  = MI_at_50  / total_mutual_information;
frac_100 = MI_at_100 / total_mutual_information;

fprintf('proportion of mutual information after latency (50ms):  %.3f bits (%.1f%%
of total)\n', MI_at_50,  frac_50*100);
```

```
proportion of mutual information after latency (50ms):  0.502 bits (70.0% of total)
```

```
fprintf('proportion of mutual information after latency (100ms): %.3f bits (%.1f%%
of total)\n', MI_at_100, frac_100*100);
```

```
proportion of mutual information after latency (100ms): 0.616 bits (85.8% of total)
```

```
figure;
fill([time_ms, fliplr(time_ms)], ...
     [MI_low', fliplr(MI_high')], ...
     [0 0 0], 'EdgeColor', 'none', 'FaceAlpha', 0.5);
hold on;
plot(time_ms, mutual_information, 'k-', 'LineWidth', 1.5);
yline(threshold,    'b--', 'Threshold',          'LabelVerticalAlignment','bottom');
xline(latency_ms,   'r--', sprintf('Latency: %d ms', latency_ms));
xline(time_ms(bin_50),  'r--', '+50ms');
xline(time_ms(bin_100), 'r--', '+100ms');

% Annotate fractions
text(time_ms(bin_50)+4,  MI_at_50,  sprintf('%.1f%%', frac_50*100),
'Color','r','FontSize',9);
text(time_ms(bin_100)+4, MI_at_100, sprintf('%.1f%%', frac_100*100),
'Color','r','FontSize',9);

xlabel('Time (ms)');
ylabel('MI (bits)');
title('Mutual Information with Bootstrap');
xlim([0 512]);
legend('95% Bootstrapped Confidence Interval', 'Mutual Information', 'Location',
'best');
```


Mutual Information with Bootstrap
Latency: 80 ms
+50ms
+100ms
70.0%
85.8%
95% Bootstrapped Confidence Interval
Mutual Information
Threshold
MI (bits)
Time (ms)
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
-0.1
0
100
200
300
400
500